KADI SARVA VISHWAVIDHYALAYA

B.E. Semester VIII EXAMINATION (April - 2025)

Subject Code: CT803B-N **Subject Name: Neural Network and Deep Learning**
Date: 11/04/2025 **Time: 12:30 PM TO 03:30 PM** **Total Marks: 70 Marks**

Instructions:
1. **All questions are compulsory.**
2. **Figures to the right indicate full marks.**
3. **Use of scientific calculator is permitted.**
4. **Indicate clearly, the options you attempt along with its respective question number.**
5. **Use the last page of main supplementary for rough work.**

Section-I

Q:1 Attempt Following.

(A) What is machine learning? And, what is neural Networks? [5]

(B) Define: [5]
1. Neuron
2. weights
3. bias
4. activation function.

(C) Types of Neural Network in Machine Learning. How Does a Neural Network work? [5]

OR

Q:1 (C) What is Linear Perceptron? Explain Perceptron Learning Algorithm and Linear separability. [5]

Q:2 Answer the following question.

(A) What is Backpropagation? How Backpropagation Works? [5]

(B) Explain Empirical Risk Minimization (ERM). [5]

OR

Q:2 (A) Explain Multilayer Perceptron? [5]

(B) Explain in detail: Gradient Descent algorithm. [5]

Q:3 Answer the following question.

(A) How Does a Computer Read an Image? [5]

(B) What Are Convolutional Neural Networks? [5]

OR

Q:3 (A) Explain ImageNet classification. [5]

(B) Difference between LeNet and AlexNet. [5]

Section II

Q:4 (All Compulsory)

(A) What Are Recurrent Neural Networks? [5]

(B) Explain Deep Learning and discuss Deep Nets and Shallow Nets. [5]

(C) Explain Training Recurrent Neural Networks? [5]

OR

Q:4 (C) Explain Difficulty of training deep neural networks. [5]

Q:5 Answer the following question.

(A) Explain Restricted Boltzman Networks or Autoencoders – RBNs and Deep Belief Networks – DBNs and Generative Adversarial Networks – GANs. [5]

(B) Explain Training of Restricted Boltzmann Machine. [5]

OR

Q:5 (A) What are Generative Models? [5]

(B) What Are Generative Adversarial Network? [5]

Q:6 Answer the following question.

(A) How To Train a GAN? [5]

(B) Challenges Of Generative Adversarial Network and Generative Adversarial Network Applications. [5]

OR

Q:6 (A) Explain MCMC and Gibbs Sampling. [5]

(B) Difference between Autoencoders & RBMs. [5]